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THEORETICAL NEUROSCIENCE

TD7: RATE MODELS

All TD materials will be made available at `https://github.com/yfardella/Theo_Neuro_TD_26_27`.

This tutorial aims to study a very standard example of rate model: the Ring Model. This model was originally introduced for orientation coding in the visual cortex. More generally, this model could apply for any network encoding a circular one-dimensional variable. In order to understand the model's dynamics, we will use methods used in the previous tutorials to study nonlinear ODEs and introduce perturbation theory.

1 Nullclines, flow and fixed point

We analyze a simple model in which all of the excitatory neurons are described by a single firing rate ν_E and all of the inhibitory neurons by a second firing rate ν_I. The firing rates obey

$$\tau_E \frac{d\nu_E}{dt}(t) = -\nu_E + \text{ReLU}\left(M_{EE}\nu_E + M_{EI}\nu_I - \gamma_E\right), \tag{1}$$

$$\tau_I \frac{d\nu_I}{dt}(t) = -\nu_I + \text{ReLU}\left(M_{IE}\nu_E + M_{II}\nu_I - \gamma_I\right), \tag{2}$$

where $\text{ReLU}(\cdots)$ is the linear threshold function (ReLU), M_{ab} is the strength of the connection *from* population b *onto* population a, and γ_a is the threshold of population a.

In this tutorial, unless stated otherwise, we use $M_{EE} = 1.25$, $M_{EI} = -1$, $M_{IE} = 1$, $M_{II} = 0$, $\gamma_E = -10$ Hz, $\gamma_I = 10$ Hz and $\tau_E = 10$ ms, with τ_I varying. Rates are in Hz and time constants in ms.

1. Draw the nullclines for ν_E and ν_I. What is the (qualitative) condition for the existence of a fixed point?

2. Draw the flow field of ν_E and ν_I above and below the nullclines (draw only the vertical and horizontal arrows).

3. Compute the fixed point (ν_E^*, ν_I^*) and check that the arguments of both ReLUs are positive there, so that the computation of the nullclines above is self-consistent.

4. Let τ_I change. Does the position of the fixed point change? Do the nullclines change? Why does this matter for the rest of the tutorial?

2 Linear stability analysis

Let $\delta\boldsymbol{\nu} = (\nu_E - \nu_E^*, \nu_I - \nu_I^*)^T$. Near the fixed point (both ReLU arguments positive) the dynamics are linear: $\dot{\delta\boldsymbol{\nu}} = A\,\delta\boldsymbol{\nu}$.

5. Write the Jacobian matrix A. Express its trace T and determinant D as a function of the M_{ab}, τ_E and τ_I, and give their numerical values.

6. Study the stability of the fixed point. What happens if the fixed point is not stable to perturbations?
   Hint: Use the values $\tau_I = 30$ ms, 40 ms and 50 ms.

7. Find the critical value τ_I^c at which the fixed point loses stability for general M_{EE}, M_{II} and τ_E, and the angular frequency ω_c of the oscillation at the bifurcation. Compute them for the parameters of the tutorial. What does the result tell you about the relative speed of excitation and inhibition needed for oscillations?

8. For $\tau_I = 30$ ms, how long does a small perturbation take to decay (time constant), and how many oscillation periods are visible before it has vanished?

9. **Why $M_{EE} > 1$ is needed.** Show that if $M_{EE} < 1$ and $M_{II} < 1$ the fixed point is stable for every τ_E, τ_I (assuming the linear regime). What does this imply for the origin of the oscillations?

3 Beyond linear stability: the limit cycle

10. Linear theory predicts exponentially growing oscillations for $\tau_I > 40$ ms. Why can this not be the final behavior? Which assumption of the linear analysis breaks down?

11. Show that the model is *piecewise linear*. Write the equations when (a) ν_E is silent and (b) ν_I is silent and interpret each of them.

4 Bonus question

12. **Inhibitory self-coupling.** Take $M_{II} = -0.5$ (self-inhibition of I) and keep the other parameters. Recompute τ_I^c and the fixed point. Does self-inhibition favor or hinder oscillations?